

Infant Mortality and Household Wealth in Nigeria

An Analysis of DHS Birth History Data

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Abstract

Using birth history data from the Nigeria Demographic and Health Survey, I reshape woman-level records into a child-level panel of 19,639 births across 239 survey clusters. I construct an infant mortality indicator, handle missing age-at-death data through hot-deck imputation, and estimate the association between household wealth and infant death. I find an infant mortality rate of 102.6 per 1,000 births and a negative association between wealth and mortality ($\beta = -0.025$, $p < 0.001$). Linking cluster GPS coordinates to WorldClim temperature data reveals that villages in hotter parts of Nigeria have significantly higher infant mortality. The full analysis code is available as a companion R script.

Data Sources

- **WomanData.csv** – Nigeria DHS women’s birth history records (6,344 women). [Download link](#).
- **Locations.csv** – GPS coordinates for 239 survey clusters. [Download link](#).
- **WorldClim 2.1** – Climatological average temperature at 10-minute resolution (June layer). Available at <https://www.worldclim.org/data/worldclim21.html>.

Reproducibility

The companion R script ([Singla.code.R](#)) reproduces every number and figure in this document. To re-run the analysis, place the following files in a single directory:

1. [Singla.code.R](#) (the R script)
2. [WomanData.csv](#) (DHS birth history data)
3. [Locations.csv](#) (cluster GPS coordinates)
4. [wc2.1_10m_tavg_06.tif](#) (WorldClim June temperature raster, downloaded and unzipped from the link above)

Then open R in that directory and run the script. It uses no hardcoded paths, so all reads and writes happen relative to the current directory. The script will produce three PDF figures ([figure1.pdf](#), [figure2.pdf](#), [figure3.pdf](#)) and print all summary statistics to the console.

1 Data Preparation

The birth history sits in wide format, with each variable (survival status, age at death, birth order, etc.) repeated 20 times per woman. I reshaped this into a child-level dataset using `pivot_longer`, splitting each column name at the final underscore to recover the variable name and child index. This produces $6,344 \times 20 = 126,880$ rows, of which 19,639 are actual births (identified by a non-missing birth index). The remaining 107,241 rows are empty slots for women with fewer than 20 children and are dropped.

One complication was that R read some birth history columns as character and others as numeric, depending on what values appeared in each column. This caused `pivot_longer` to fail when trying to stack them. I resolved it by casting all birth columns to character before reshaping and converting the numeric ones back afterward.

Table 1: Data after reshaping

	Count
Women in original data	6,344
Valid births retained	19,639
of which alive at interview	16,497
of which dead	3,142
Survey clusters	239

2 Constructing the Infant Mortality Indicator

I define infant mortality as death at or before 12 completed months of age. The DHS variable `b5` records whether the child was alive at interview, and `b7` gives age at death in completed months. A child is classified as an infant death if `b5 = 'no'` and `b7 ≤ 12`.

Twelve dead children had missing `b7` values. The DHS Recode6 manual describes a hot-deck imputation procedure for this situation: assign the child the same age at death as the last child of the same birth order encountered in the data file. The flag variable `b13` records whether DHS applied this imputation. I replicated the procedure for the 12 remaining cases, sorting by birth order and carrying forward the most recent `b7` from a dead child of the same birth order. After imputation, no dead child has a missing age at death.

The resulting infant mortality rate is 102.6 per 1,000 births (2,015 infant deaths out of 19,639 births).

3 Wealth and Infant Mortality

Individual-level regression

The DHS wealth index is a PCA-derived continuous score based on household assets, housing materials, and water/sanitation access, standardized to mean 0 and SD 1 across the national sample. I regress the infant death indicator on this index.

$$\text{infant_death}_i = \alpha + \beta \cdot \text{wealth_index}_i + \varepsilon_i$$

Table 2: OLS regression of infant death on household wealth

	Coefficient	Std. Error
Intercept	0.098	0.002
Wealth index	−0.025	0.002
Observations	19,639	
R^2	0.006	

A one standard deviation increase in household wealth is associated with about 25 fewer infant deaths per 1,000 births. The coefficient is highly significant ($t = -10.8$) but the R^2 is low, which is typical for individual-level binary outcome regressions with a single predictor.

This estimate captures correlation, not causation. Wealthier households also tend to have more educated mothers, better healthcare access, and live in different geographic areas. Any of these could independently drive both wealth and child survival. There is also a reverse causality concern, since infant deaths can reduce household wealth through medical costs and lost productivity. Credibly identifying a causal effect would require exogenous variation in wealth, such as a randomized cash transfer or a natural experiment.

Village-level patterns

Aggregating to the cluster level reveals the same pattern. Figure 1 shows that villages with higher average wealth tend to have lower infant mortality.



Figure 1: Village-average wealth versus infant mortality probability across 239 clusters.

4 Geography of Infant Mortality

Mapping cluster-level mortality rates across Nigeria reveals clear spatial variation (Figure 2). Higher mortality clusters tend to concentrate in the northern regions, consistent with known disparities in healthcare infrastructure and economic development between northern and southern Nigeria.

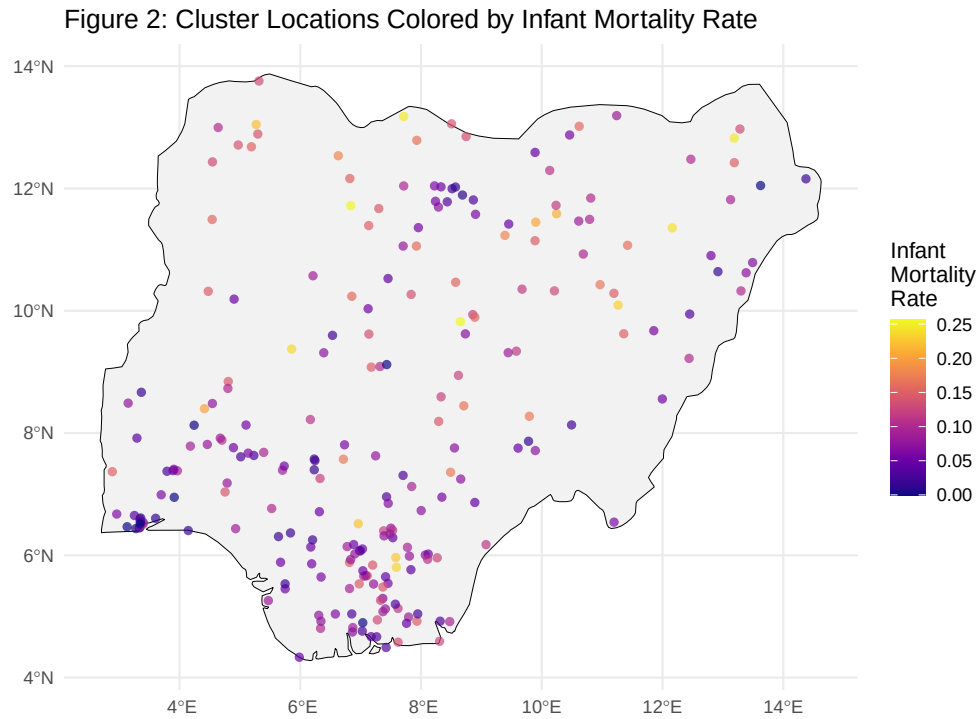


Figure 2: Survey cluster locations colored by infant mortality rate. Rates range from 0.000 to 0.256.

5 Temperature and Infant Mortality

To explore whether climate conditions are associated with infant mortality, I linked each cluster's GPS coordinates to WorldClim June average temperature data at 10-minute (~ 18 km) resolution. DHS displaces cluster coordinates by up to 2 km (urban) or 5 km (rural) for respondent privacy, but this is small relative to the grid cell size.

Table 3: Village-level regression of infant mortality on June temperature

	Coefficient	Std. Error
Intercept	-0.097	0.052
June temperature ($^{\circ}\text{C}$)	0.007	0.002
Observations	239	
R^2	0.051	

The positive coefficient ($p = 0.0004$) indicates that hotter villages have higher infant mortality. A one degree Celsius increase in average June temperature is associated with 7 additional infant deaths per 1,000 births at the village level. This is consistent with the geographic pattern in Figure 2, since northern Nigeria is both hotter and has higher mortality.

Figure 3: Average June Temperature vs. Village Mortality Rate

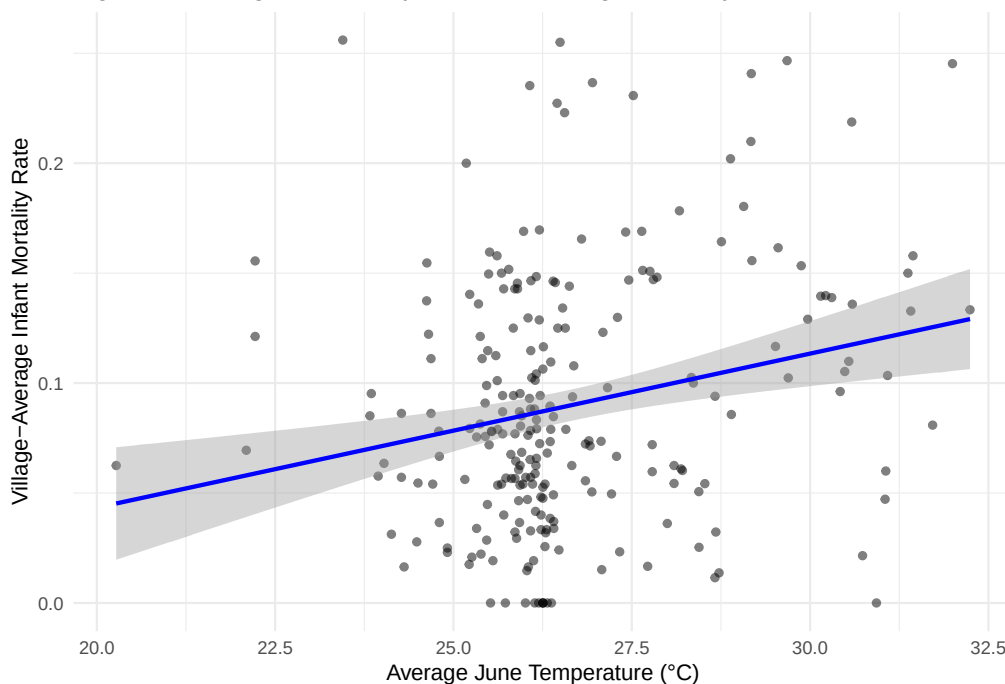


Figure 3: June temperature versus village infant mortality rate with OLS line and 95% confidence band.

Of course, temperature here is likely proxying for a bundle of geographic, economic, and institutional factors that differ between northern and southern Nigeria. The regression does not isolate the causal effect of temperature itself.